

CpG Island Tumor Suppressor Promoter Methylation in Non-BRCA-Associated Early Mammary Carcinogenesis

Background: Only 5% of all breast cancers are the result of BRCA1/2 mutations. Methylation silencing of tumor suppressor genes is well described in sporadic breast cancer; however, its role in familial breast cancer is not known. Methods: CpG island promoter methylation was tested in the initial random periareolar fine-needle aspiration sample from 109 asymptomatic women at high risk for breast cancer. Promoter methylation targets included RARB (M3 and M4), ESR1, INK4a/ARF, BRCA1, PRA, PRB, RASSF1A, HIN-1, and CRBP1. Results: Although the overall frequency of CpG island promoter methylation events increased with age ($P < 0.0001$), no specific methylation event was associated with age. In contrast, CpG island methylation of RARB M4 ($P = 0.051$), INK4a/ARF ($P = 0.042$), HIN-1 ($P = 0.044$), and PRA ($P = 0.032$), as well as the overall frequency of methylation events ($P = 0.004$), was associated with abnormal Masood cytology. The association between promoter methylation and familial breast cancer was tested in 40 unaffected premenopausal women in our cohort who underwent BRCA1/2 mutation testing. Women with BRCA1/2 mutations had a low frequency of CpG island promoter methylation (15 of 15 women had ≤ 4 methylation events), whereas women without a mutation showed a high frequency of promoter methylation events (24 of 25 women had 5-8 methylation events; $P < 0.0001$). Of women with a BRCA1/2 mutation, none showed methylation of HIN-1 and only 1 of 15 women showed CpG island methylation of RARB M4, INK4a/ARF, or PRB promoters. Conclusions: This is the first evidence of CpG island methylation of tumor suppressor gene promoters in non-BRCA1/2 familial breast cancer.